Milan J. Hancock

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Education

Indiana University Bloomington - Luddy School of Informatics, Computing, and Engineering

B.S. Intelligent Systems Engineering *(Expected May 2028)*

Specialization: Cyber-Physical / Computer Engineering

Dean's List: Fall 2024 - Spring 2026 | 3.99/4.00 GPA

Skills

Programming & HDL: C/C++, Python, SystemVerilog

Hardware & Embedded Systems: FPGA Development (Xilinx), Arduino, Raspberry Pi, Circuit Prototyping, Soldering

Engineering & Design Tools: Vivado, LTspice, Fusion 360, Onshape, Git, Linux

Machine Learning: scikit-learn, TensorFlow, Google Colab

Projects

Real-Time Driving Event Classification Using BeamNG.tech | Independent Project, In Progress

- Developing a Python-based data collection pipeline using BeamNG.tech to capture simulated vehicle accelerometer and gyroscope measurements.
- Investigating IMU-based detection of driving events, including hard braking, rapid acceleration, and swerving.
- Planning supervised classification and sliding-window inference to evaluate the feasibility of real-time driving-event detection.

FPGA Digital Design — EMA Filter and Population Count | SystemVerilog, Xilinx PYNQ

- Collaborated with a partner to implement exponential moving average (EMA) filtering and population count (popcount) logic in SystemVerilog.
- Simulated and verified module functionality using provided testbenches.
- Implemented and tested both designs on a Xilinx PYNQ FPGA development board.

DIY Sim Racing Shifter and Handbrake | Arduino Pro Micro, C++, CAD, 3D Printing

- Designed and built a custom USB sequential shifter and analog handbrake using a shared Arduino Pro Micro.
- Integrated microswitches and a repurposed analog joystick sensor with DIY mechanical linkages to capture user inputs.
- Developed C++ firmware to process digital and analog inputs and calibrate the handbrake's usable axis range.
- Designed and 3D-printed mechanical components and assembled adjustable linkages for reliable operation.

Activities

Aerial Robotics Club (Fall 2025)

- Participated in team drone design, Arduino integration, and iterative prototype development.

Work Experience

Cascades Golf Course: Laborer *(May 2025 - Present)*

TurfPro Plus: Groundskeeper *(May 2024 - August 2024)*

Sam's Club: Member Assist *(May 2023 - August 2023)*